

SECR2043 OPERATING SYSTEMS

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 Section : 03

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write **"No output"** instead.

1.	Display your current working directory.	
	Command	Output
	pwd	<pre>adriana@secr2043:~\$ pwd /home/adriana adriana@secr2043:~\$</pre>
2.	List all the files and directories in your home directory, including hidden ones.	
	Command	Output
	ls -a	<pre>adriana@secr2043:~\$ ls -a . .local filelist.txt .. .profile 'first nano' .bash_history .ssh nano.1191.save .bash_logout .sudo_as_admin_successful practice.txt .bashrc combined.txt sample.txt .cache file2.txt workspace adriana@secr2043:~\$</pre>
3.	Create a new directory named "os_lab" in your home directory.	
	Command	Output
	mkdir os_lab ls	<pre>adriana@secr2043:~\$ mkdir os_lab adriana@secr2043:~\$ ls combined.txt 'first nano' practice.txt file2.txt nano.1191.save sample.txt filelist.txt os_lab workspace adriana@secr2043:~\$</pre>
4.	Change your current working directory to "os_lab".	
	Command	Output
	cd os_lab	<pre>adriana@secr2043:~\$ cd os_lab adriana@secr2043:~/os_lab\$ </pre>
5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.	
	Command	Output
	touch hello.txt ls echo "Hello, world!" > hello.txt	<pre>adriana@secr2043:~/os_lab\$ touch hello.txt adriana@secr2043:~/os_lab\$ ls hello.txt adriana@secr2043:~/os_lab\$ echo "Hello, world!" > hello.txt adriana@secr2043:~/os_lab\$</pre>

6.	Display the content of "hello.txt".	
	Command	Output
	cat hello.txt	<pre>adriana@secr2043:~/os_lab\$ cat hello.txt Hello, world! adriana@secr2043:~/os_lab\$</pre>
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.	
	Command	Output
	cp hello.txt hello_copy.txt cat hello_copy.txt	<pre>adriana@secr2043:~/os_lab\$ cp hello.txt hello_copy.txt adriana@secr2043:~/os_lab\$ cat hello_copy.txt Hello, world! adriana@secr2043:~/os_lab\$</pre>
8.	Rename "hello_copy.txt" to "hello_bak.txt".	
	Command	Output
	mv hello_copy.txt hello_bak.txt cat hello_bak.txt	<pre>adriana@secr2043:~/os_lab\$ mv hello_copy.txt hello_bak.txt adriana@secr2043:~/os_lab\$ cat hello_bak.txt Hello, world! adriana@secr2043:~/os_lab\$</pre>
9.	Delete "hello.txt".	
	Command	Output
	rm hello.txt ls	<pre>adriana@secr2043:~/os_lab\$ rm hello.txt adriana@secr2043:~/os_lab\$ ls hello_bak.txt</pre>
10.	Display your home directory with additional information, together with subdirectories contents.	
	Command	Output
	cd ls -R	<pre>adriana@secr2043:~/os_lab\$ cd adriana@secr2043:~\$ ls -R .: combined.txt 'first nano' practice.txt file2.txt nano.1191.save sample.txt filelist.txt os_lab workspace ./os_lab: hello_bak.txt ./workspace: argument hello_world.c input_keyboard kernel_info.c argument.c hello_world.cpp input_keyboard.c multiply hello_world hello_world2 kernel_info multiply.c adriana@secr2043:~\$</pre>
	Total Mark:	

ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

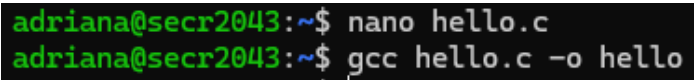
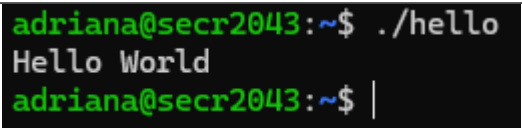
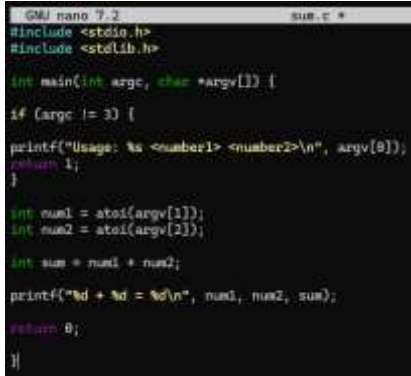
For each question, write down the command and the output (if any) in the answer sheet.

1.	Create an empty file named "hello.txt" in the current working directory.	
	Command	Output
	touch hello.txt ls	<pre>adriana@secr2043:~\$ touch hello.txt adriana@secr2043:~\$ ls combined.txt 'first nano' os_lab workspace file2.txt hello.txt practice.txt filelist.txt nano.1191.save sample.txt adriana@secr2043:~\$ </pre>
2.	Write the text "Hello World" to "hello.txt" using echo.	
	Command	Output
	echo "Hello World!" > hello.txt cat hello.txt	<pre>adriana@secr2043:~\$ echo "Hello World!" > hello.txt adriana@secr2043:~\$ cat hello.txt Hello World! adriana@secr2043:~\$</pre>
3.	Append the text "This is a test" to "hello.txt" using echo.	
	Command	Output
	echo "This is a test" >> hello.txt cat hello.txt	<pre>adriana@secr2043:~\$ echo "This is a test" >> hello.txt</pre>
4.	Display the contents of "hello.txt" using cat.	
	Command	Output
	cat hello.txt	<pre>adriana@secr2043:~\$ cat hello.txt Hello World! This is a test adriana@secr2043:~\$</pre>
5.	Create another file named "bye.txt" with the text "Goodbye World" using echo.	
	Command	Output
	touch bye.txt echo "Goodbye World" > bye.txt	<pre>adriana@secr2043:~\$ touch bye.txt adriana@secr2043:~\$ echo "Goodbye World" > bye.txt adriana@secr2043:~\$ cat bye.txt Goodbye World adriana@secr2043:~\$</pre>
6.	Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.	
	Command	Output
	cat hello.txt bye.txt	<pre>adriana@secr2043:~\$ cat hello.txt bye.txt Hello World! This is a test Goodbye World adriana@secr2043:~\$</pre>
7.	Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.	
	Command	Output
	cat hello.txt bye.txt > greeting.txt	<pre>adriana@secr2043:~\$ cat hello.txt bye.txt > greeting.txt</pre>

8.	Display the contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	<pre> adriana@secr2043:~\$ cat greeting.txt Hello World! This is a test Goodbye World adriana@secr2043:~\$ </pre>
9.	Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.	
	Command	Output
	nano greeting.txt	<pre> GNU nano 7.2 greeting.txt * Hello Adriana! This is a test Goodbye Adriana </pre>
10.	Display the modified contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	<pre> adriana@secr2043:~\$ cat greeting.txt Hello Adriana! This is a test Goodbye Adriana </pre>
	Total Mark:	

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc. For each question, write down the command and the output (if any) in the answer sheet.

1.	Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".	
	Command	Output
	nano hello.c	
2.	Compile the program "hello.c" using gcc and generate an executable file named "hello".	
	Command	Output
	gcc hello.c -o hello	
3.	Run the executable file "hello" and display its output.	
	Command	Output
	./hello	
4.	Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".	
	Command	Output
	nano sum.c	
5.	Compile the program "sum.c" using gcc and generate an executable file named "sum".	
	Command	Output
	gcc sum.c -o sum	
6.	Run the executable file "sum" with 10 and 20 as arguments and display its output.	
	Command	Output
	./sum 10 20	

7.	Run the executable file "sum" with -5 and 15 as arguments and display its output.	
	Command	Output
	./sum -5 15	<pre> adriana@secr2043:~\$./sum -5 15 -5 + 15 = 10 adriana@secr2043:~\$ </pre>
8.	Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".	
	Command	Output
	nano echo.c	<pre> GNU nano 7.2 echo.c #include <stdio.h> int main() { char input[1000]; printf("Enter a line of text: "); if (fgets(input, sizeof(input), stdin) != NULL) { printf("You entered: %s", input); } else { printf("Error reading input.\n"); } return 0; } </pre>
9.	Compile the program "echo.c" using gcc and generate an executable file named "echo".	
	Command	Output
	gcc echo.c -o echo	<pre> adriana@secr2043:~\$ nano echo.c adriana@secr2043:~\$ gcc echo.c -o echo adriana@secr2043:~\$ </pre>
10.	Run the executable file "echo", type "Hello World" as input and display its output.	
	Command	Output
	./echo	<pre> adriana@secr2043:~\$./echo Enter a line of text: Hello World You entered: Hello World adriana@secr2043:~\$ </pre>
	Total Mark:	